Rahul Suryakant Sakhare, Ph.D., P.E.

Curriculum Vitae

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EDUCATION

- **Doctor of Philosophy (Ph.D.)** — May 2023
  Purdue University, West Lafayette, United States
  Civil Engineering — Transportation & Infrastructure Systems
- **Master of Technology (M.Tech.)** — May 2018
  Indian Institute of Technology (IIT) Madras, India
  Civil Engineering (Transportation Engineering)
- **Bachelor of Technology (B.Tech.)** — May 2017
  Indian Institute of Technology (IIT) Madras, India
  Civil Engineering (Minor: Management Studies)

PROFESSIONAL REGISTRATION

- **Professional Engineer (PE)**, Civil – State of Indiana, License No. PE12600619 — Jun 2026
- Google Cloud Certified **Generative AI Leader** — May 2026

PROFESSIONAL APPOINTMENTS

- **Transportation Research Engineer** (Post-doctoral equivalent) — Aug 2023 – Present
  Joint Transportation Research Program (JTRP)
  Lyles School of Civil & Construction Engineering
  Purdue University, West Lafayette, United States
- **Graduate Research Assistant** — Jan 2020 – May 2023
  Advisor: Prof. Darcy M. Bullock
  Lyles School of Civil Engineering
  Purdue University, West Lafayette, United States
- **Executive Consultant, Transportation Team** (Industry Experience) — Jul 2018 – Nov 2019
  Ernst & Young (EY) LLP, New Delhi, India
- **Visiting Undergraduate Research Student** — May – Jul 2017
  Purdue University, West Lafayette, United States
- **Summer Researcher — Purdue Undergraduate Research Experience (PURE)** — May – Jul 2016
  Lyles School of Civil Engineering, Purdue University, United States

RESEARCH INTERESTS

Transportation Systems Engineering, Connected and Autonomous Vehicles (CAVs), Cloud-Native Big Data Analytics, Traffic Safety and Work Zone Operations, Applied AI, Network Performance Monitoring & Infrastructure Resilience, Cyber-Physical Transportation Systems.

SCIENTIFIC PUBLICATIONS

Google Scholar citations: **521** (Sept. 28, 2026); h-index 13 and i10-index 19

Theses

- **PhD Dissertation:** Integrating Connected Vehicle Data for Operational Decision Making (https://doi.org/10.25394/PGS.22684822.v1) Purdue University, April 2023.
  Committee: Dr. Darcy Bullock, Dr. Samuel Labi, Dr. Konstantina Gkritza, Dr. James Krogmeier
- **Master's Thesis:** Reliable Corridor Level Travel Time Estimation using Probe Vehicle Data. Indian Institute of Technology (IIT) Madras, Chennai, May 2018.
  Committee: Dr. V. Lelitha Devi, Dr. Gitakrishnan Ramadurai, Dr. Atul Narayan, Dr. A. Veeraragavan

Monograph & Patent

- **Sakhare, R. S.**, Desai, J., Mathew, J. K., McGregor, J., Kachler, M., & Bullock, D. M. (2024). Measuring and visualizing freeway traffic conditions: Using connected vehicle data (Joint Transportation Research Program No. TPF-5(514)). West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284317751 (528 downloads as of Sept. 28, 2026)
- (Patent) Devi, L. & **Sakhare, R. S.** (2024). System and Method for Estimating Reliable Corridor Level Travel Time Estimation Using Probe Vehicle Data (India Patent No. 529133). *The Patent Office, Government of India*.

Peer-Reviewed Journal Publications

J32. **Sakhare, R.S.**, Desai, J., Gartner, C., Shellhamer, N., Ruble, T., McCoy, D., & Bullock, D.M. (2026). Deriving systemwide granular segment level traffic mobility performance metrics using connected vehicle data to identify improvement opportunities. ***Transportation***. https://doi.org/10.1007/s11116-026-10736-8

J31. **Sakhare, R.S.**, Desai, J., Overall, M., Mukai, J., Pava, J., McGregor, J., & Bullock, D.M. (2026). Work Zone Performance Measures Derived from Connected Vehicle Data for Safety and Mobility Assessment. *Future Transportation,* 6(1), 12. https://doi.org/10.3390/futuretransp6010012

J30. Desai J.C., Gartner, C.M., **Sakhare, R.S.**, Cox E. & Bullock, D.M. (2026). Evaluating Safety Benefits of Ramp Metering by Leveraging Connected Vehicle Data. ***IEEE Open Journal of Intelligent Transportation Systems***, 7, 190-199. https://doi.org/10.1109/OJITS.2025.3650100

J29. Mukai, J., Desai, J., **Sakhare, R.S.**, Sturdevant, N. and Bullock, D. (2026) Quantifying Hard Breaking Events Occurring Prior to Secondary Crashes on Indiana Interstates. *Journal of Transportation Technologies*, 16, 150-165. https://doi.org/10.4236/jtts.2026.161010

J28. Overall, M., Mukai, J., **Sakhare, R.S.**, Desai, J., Horton, D. and Bullock, D.M. (2026). Applications of Commercial Truck Dash Cameras for Work Zone Inspection and Monitoring. *Journal of Transportation Technologies*, 16, 95-114. https://doi.org/10.4236/jtts.2026.161006

J27. Gartner, C.M., **Sakhare, R.S.**, Desai J.C., Sturdevant J. & Bullock, D.M. (2025). A Scalable Data Model for Signalized Intersection Performance Measures. ***IEEE Access***, 13, 197851-197863. https://doi.org/10.1109/ACCESS.2025.3635007

J26. Desai, J., **Sakhare, R. S.**, McGregor, J., Snyder, B., & Bullock, D. M. (2025). Evaluating Impact of Automated Work Site Speed Enforcement Using Connected Vehicle Data. *Applied Sciences*, 15(22), 12140. https://doi.org/10.3390/app152212140

J25. Mathew, J., Desai, J., **Sakhare, R. S.**, Hunter, J. and Bullock, D. (2025) Spatiotemporal Analysis of Pavement Roughness Using Connected Vehicle Data for Asset Management. *Journal of Transportation Technologies, 15*, 1-16. https://doi.org/10.4236/jtts.2025.151001

J24. Overall, M. W., Mukai, J., **Sakhare, R. S.**, Desai, J., Lowther, H., & Bullock, D. M. (2025). Measuring the Impact of Recovery Resource Delay on Traffic Incident Management Clearance Times. *Future Transportation*, 5(4), 171. https://doi.org/10.3390/futuretransp5040171

J23. Driscoll, T., **Sakhare, R.S.**, Desai, J. and Bullock, D.M. (2025) Utilizing Connected Vehicle Data to Identify Impacts of Congestion on Adjacent Agencies. *Journal of Transportation Technologies*, 15, 522-541. https://doi.org/10.4236/jtts.2025.154024

J22. **Sakhare, R.S.**, Desai, J., Mathew, J.K. and Bullock, D.M. (2024). Assessing the Interstate Coverage of Commercial Trucks Capable of Providing Roadway Imagery via On-Vehicle Dash Camera in the United States. ***IEEE Access****, vol. 12*, pp. 173517-173529. https://doi.org/10.1109/ACCESS.2024.3503368

J21. **Sakhare, R.S.**, Desai, J., Mathew, J. and Bullock, D.M. (2024). Impact of 2024 Solar Eclipse on National Traffic Mobility using Connected Vehicle Data and Images. ***Transportation Research Interdisciplinary Perspectives***, 27. https://doi.org/10.1016/j.trip.2024.101225

J20. Brinster, G. L., Desai, J., **Sakhare, R. S.**, Mathew, J. K., & Bullock, D. M. (2024). Expanding on methodologies for analyzing truck speeds in states with differential speed limits. *Safety,* 10(4), 99. https://doi.org/10.3390/safety10040099

J19. Vajpayee, V., Saldivar-Carranza, E. D., **Sakhare, R. S.**, & Bullock, D. M. (2024). Large Scale Evaluation of Normalized Hard-Braking Events Derived from Connected Vehicle Trajectory Data at Signalized Intersections, Roundabouts, and All-Way Stops. *Future Transportation, 4*(3), 968-984. https://doi.org/10.3390/futuretransp4030046

J18. Brinster, G.L., Desai, J., Overall, M.W., Gartner, C., **Sakhare, R.S.**, Mathew, J.K., Evans, N. and Bullock, D. (2024) Evaluating the Robustness of MDSS Maintenance Forecasts Using Connected Vehicle Data. *Journal of Transportation Technologies*, 14, 549-569. https://doi.org/10.4236/jtts.2024.144030

J17. Saldivar-Carranza, E. D., **Sakhare, R. S.**, Desai, J., Mathew, J. K., Sivakumar, A. J., Mukai, J., & Bullock, D. M. (2024). Impact of Privacy Filters and Fleet Changes on Connected Vehicle Trajectory Datasets for Intersection and Freeway Use Cases. *Smart Cities, 7*(5), 2366-2391. https://doi.org/10.3390/smartcities7050093

J16. Desai, J., Saldivar-Carranza, E. D., **Sakhare, R. S.**, Mathew, J. K., & Bullock, D. M. (2024). Impact of In-Cab Alerts on Connected Truck Speed Reductions in Indiana. *Vehicles, 6*(4), 1857-1871. https://doi.org/10.3390/vehicles6040090

J15. **Sakhare, R. S.**, Desai, J., Saldivar, E. D., & Bullock, D. M. (2023). Methodology for Monitoring Border Crossing Delays with Connected Vehicle Data: United States and Mexico Land Crossings Case Study. *Future Transportation,* 4(1), 107-129. https://doi.org/10.3390/futuretransp4010007

J14. **Sakhare, R. S.**, Li, H., & Bullock, D. M. (2023). Methodology for the Identification of Shock Wave Type and Speed in a Traffic Stream Using Connected Vehicle Data. *Future Transportation,* 3(4), 1147–1174. https://doi.org/10.3390/futuretransp3040063

J13. **Sakhare, R. S.**, Zhang, Y., Li, H., & Bullock, D. M. (2023). Impact of Rain Intensity on Interstate Traffic Speeds Using Connected Vehicle Data. *Vehicles,* 5(1), 133–155. https://doi.org/10.3390/vehicles5010009

J12. Desai, J., Mathew, J. K., Li, H., **Sakhare, R. S.**, Horton, D., & Bullock, D. M. (2023). Analysis of Connected Vehicle Data to Quantify National Mobility Impacts of Winter Storms for Decision Makers and Media Reports. Future Transportation, 3(4), 1292-1309. https://doi.org/10.3390/futuretransp3040071

J11. Desai, J., Mathew, J., Li, H., Mukai, J., **Sakhare, R. S.**, & Bullock, D. M. (2023). Correlating Connected Vehicle Estimated Deceleration Events with Crash Incidents near Interstate Interchanges. *Journal of Transportation Technologies*, 13, 674-688. https://doi.org/10.4236/jtts.2023.134031

J10. **Sakhare, R. S.**, Hunter, M., Mukai, J., Li, H., & Bullock, D. M. (2022). Truck and Passenger Car Connected Vehicle Penetration on Indiana Roadways. *Journal of Transportation Technologies,* 12, 578-599. https://doi.org/10.4236/jtts.2022.124034

J9. **Sakhare, R. S.**, Desai, J., Li, H., Kachler, M. A., Bullock, D. M. (2022). Methodology for Monitoring Work Zones Traffic Operations Using Connected Vehicle Data. *Safety,* 8(2), 41. **(Editor's Choice Award)** https://doi.org/10.3390/safety8020041

J8. Mahlberg, J. A., Desai, J., Li, H., **Sakhare, R. S.**, Wells, T., Bullock, D. M. (2022). Calibration and Confidence in Snowplow Fleet Operations and Fleet Telematics. *Journal of Transportation Technologies*, 12(4), 696-710. https://doi.org/10.4236/jtts.2022.124040

J7. **Sakhare, R. S.**, Desai, J. C., Mahlberg, J., Mathew, J. K., Kim, W., Li, H., McGregor, D., & Bullock, D. M. (2021). Evaluation of the Impact of Queue Trucks with Navigation Alerts Using Connected Vehicle Data. *Journal of Transportation Technologies*, 11(4), 561-576. https://doi.org/10.4236/jtts.2021.114035 **(Recognized by ITS Midwest as the 2021 Project of the Year)**

J6. **Sakhare, R. S.**, Desai, J. C., Mathew, J. K., McGregor, J. D., & Bullock, D. M. (2021). Evaluation of the impact of presence lighting and digital speed limit trailers on interstate speeds in Indiana work zones. *Journal of transportation technologies*, 11(2), 157-167. https://doi.org/10.4236/jtts.2021.112010

J5. Mathew, J. K., Desai, J. C., **Sakhare, R. S.**, Kim, W., Li, H., & Bullock, D. M. (2021). Big data applications for managing roadways. Institute of Transportation Engineers. ITE Journal, 91(2), 28-35.

J4. Mahlberg, J. A., **Sakhare, R. S.**, Li, H., Mathew, J. K., Bullock, D. M., & Surnilla, G. C. (2021). Prioritizing Roadway Pavement Marking Maintenance Using Lane Keep Assist Sensor Data. *Sensors*, 21(18), 6014. https://doi.org/10.3390/s21186014

J3. Desai, J., Mahlberg, J., Kim, W., **Sakhare, R. S.**, Li, H., McGuffey, J., & Bullock, D. M. (2021). Leveraging telematics for winter operations performance measures and tactical adjustment. *Journal of Transportation Technologies*, 11(4), 611-627. https://doi.org/10.4236/jtts.2021.114038

J2. **Sakhare, R. S.** & Vanajakshi, L. (2020). Reliable Corridor Level Travel Time Estimation Using Probe Vehicle Data. ***Transportation Letters***, 12:8, 570-579. https://doi.org/10.1080/19427867.2019.1671041

J1. **Sakhare, R. S.**, Dev, L. K., Prasad, K. E., & Robinson, R. G. (2016). A Double Acting Piston Based Automatic Volume Change Apparatus. Geotechnical Testing Journal, 39(6), 899-905. http://doi.org/10.1520/GTJ20150296 (**IGS Best Paper Award**)

Key Project / Technical Reports

T22. Thompson, A., Desai, J. C., **Sakhare, R. S.**, & Bullock, D. M. (2026). Research support to INDOT on systemwide asset condition assessment using connected vehicle data (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2026/04). West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284318612

T21. **Sakhare, R. S.**, Desai, J., Gartner, C., Mukai, J., Sturdevant, N., Shellhamer, N., Ruble, T., McCoy, D., & Bullock, D. M. (2025). Connected vehicle trajectory data to screen network for hard-braking and hard-acceleration events (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2025/30). West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284318579

T20. Desai, J., **Sakhare, R. S.**, McGregor, J., Sturdevant, J., & Bullock, D. M. (2025). Worksite speed control system (WSCS) (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2025/33). West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284318595

T19. Overall, M. W., Mukai, J., Desai, J., **Sakhare, R. S.**, Kachler, M., McGregor, J., & Bullock, D. M. (2025). Continued deployment of Indiana work zone analytics (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2025/31). West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284318581

T18. Desai, J. C., Gartner, C., **Sakhare, R. S.**, Cox, E. D., Sturdevant, J., & Bullock, D. M. (2025). Research support to INDOT on I-465 Southeast variable speed limit and ramp meter project (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2025/32). West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284318592

T17. Overall, M. W., Mukai, J., **Sakhare, R. S.**, Desai, J., Horton, D., & Bullock, D. M. (2025). Case studies documenting lessons learned and identifying opportunities for maintenance of traffic (MOT) improvements (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2025/29). West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284318578

T16. Desai, J., **Sakhare, R. S.**, Mathew, J. K., Sturdevant, N., Cox, E., & Bullock, D. M. (2025). Communication of fixed and mobile warnings to commercial trucks using in-cab notification (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2025/24). West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284318232

T15. Overall, M. W., Mukai, J., **Sakhare, R. S.**, Desai, J., Lowther, H., Cox, E., Sturdevant, J., & Bullock, D. M. (2025). Development of incident management performance measures database and supporting training material (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2025/21). West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284317907

T14. **Sakhare, R. S.**, Desai, J., Horton, D., & Bullock, D. M. (2025). Crawfordsville District interstate speed profiles: April 2020–March 2025. West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284317852 (1,888 downloads as of 2/26/2026)

T13. **Sakhare, R. S.**, Desai, J., Horton, D., & Bullock, D. M. (2025). *Indiana interstate speed profiles 2020–2024*. West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284317843

T12. Gayen, S., Saldivar-Carranza, E. D., **Sakhare, R. S.**, Desai, J., Mathew, J. K., Shellhamer, N., Ruble, T., McCoy, D., Gallagher, J., & Bullock, D. M. (2024). Statewide screening of signalized intersections for capacity improvements (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2024/20). West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284317755

T11. Brinster, G. L., Desai, J., Overall, M. W., Gartner, C., **Sakhare, R. S.**, Mathew, J. K., Evans, N., & Bullock, D. M. (2024). Evaluating the robustness of MDSS forecast and compliance with recommendations (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2024/31). West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284317805

T10. **Sakhare, R. S.**, Desai, J., Woker, W., Li, H., Mathew, J., Mahlberg, J., Saldivar-Carranza, E. D., Horton, D., & Bullock, D. M. (2023). Connected vehicle centric dashboards for TMC of the future (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2023/17). West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284317642

T9. Desai, J., **Sakhare, R. S.**, Mahlberg, J., Mathew, J. K., Li, H., & Bullock, D. M. (2023). Implementation of enhanced probe data (CANBUS) for tactical workzone and winter operations management (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2023/18). West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284317643

T8. Mahlberg, J. A., Desai, J., Malackowski, H., Li, H., Mathew, J. K., **Sakhare, R. S.**, Wells, T., & Bullock, D. M. (2023). Crowdsourcing/winter operations dashboard upgrade (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2023/21). West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284317652

T7. **Sakhare, R. S.**, Mahlberg, J., Mathew, J. K., Desai, J., Li, H., Bullock, D. M. (2022). Impacts to traffic behavior from queue warning truck: Current pilot project (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2022/26). West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284317448

T6. Mathew, J. K., Li, H., Desai, J., **Sakhare, R. S.**, Saldivar-Carranza, E., Hunter, M., Scholer, B., & Bullock, D. M. (2022). Integration of probe data tools into TMC operations (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2022/01). West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284317363

T5. Desai, J., Mathew, J. K., Li, H., **Sakhare, R. S.**, Horton, D., & Bullock, D. M. (2022). National mobility analysis for all interstate routes in the United States: December 2022. West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284317591

T4. Desai, J., Mathew, J. K., Li, H., **Sakhare, R. S.**, Horton, D., & Bullock, D. M. (2022). National mobility analysis for all interstate routes in the United States: August 2022. West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284317585

T3. **Sakhare, R. S.**, Li, H., Mathew, J. K., Desai, J., Horton, D., & Bullock, D. M. (2022). Indiana interstate speed profiles 2018–2022. West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284317589

T2. **Sakhare, R. S.**, Desai, J., Mathew, J. K., Kim, W., Mahlberg, J., Li, H., & Bullock, D. M. (2021). Evaluating the impact of vehicle digital communication alerts on vehicles (Joint Transportation Research Program Publication No. FHWA/IN/JTRP-2021/19). West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284317324

T1. Day, C. M., McNamara, M. L., Li, H., **Sakhare, R. S.**, Desai, J., Cox, E. D., Horton, D. K., & Bullock, D. M. (2016). 2015 Indiana mobility report and performance measure dashboards. West Lafayette, IN: Purdue University. https://doi.org/10.5703/1288284316352

Peer-Reviewed Conference & Workshop Presentations

C26. **Sakhare, R.S.**, Desai, J., Overall, M., Mukai, J., Pava, J., McGregor, J., & Bullock, D. M. (2026, January). Improving Vehicle and Worker Safety in Dynamic Work Zones – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-02289). TRB. https://annualmeeting.mytrb.org/OnlineProgram/Details/24853

C25. Desai, J., Gartner, C., **Sakhare, R.S.**, Cox, E., & Bullock, D. M. (2026, January). Improving Vehicle and Worker Safety in Dynamic Work Zones – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-02345). TRB. https://annualmeeting.mytrb.org/OnlineProgram/Details/24853

C24. **Sakhare, R.S.**, Desai, J., Gartner, C., Shellhamer, N., Ruble, T., McCoy, D., & Bullock, D. M. (2026, January). Travel Data and Networks: Collection, Modeling, and Mobility Insights – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-01332). TRB. https://annualmeeting.mytrb.org/OnlineProgram/Details/24601

C23. Gartner, C., **Sakhare, R.S.**, Desai, J., Sturdevant, J., & Bullock, D. M. (2026, January). Advanced Modeling, Optimization, and Safety for Signalized Intersections – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-02156). TRB. https://annualmeeting.mytrb.org/OnlineProgram/Details/24854

C22. Desai, J., **Sakhare, R.S.**, McGregor, J., Snyder, B., & Bullock, D. M. (2026, January). Data-Driven Approaches for Perception, Prediction, and Decision Making in Autonomous Vehicles – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-02221). TRB. https://annualmeeting.mytrb.org/OnlineProgram/Details/24522

C21. Overall, M., Mukai, J., **Sakhare, R.S.**, Desai, J., Horton, D., & Bullock, D. M. (2026, January). Modeling Traffic Speed and Network Performance Under Disruptive Events – Poster Session. In Transportation Research

Board 105th Annual Meeting (No. TRBAM-26-03560). TRB. https://annualmeeting.mytrb.org/OnlineProgram/Details/24845

C20. Driscoll, T., **Sakhare, R.S.**, Desai, J., & Bullock, D. M. (2026, January). Emerging Research Topic in Strategic Management and Organizational Excellence – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-03893). TRB. https://annualmeeting.mytrb.org/OnlineProgram/Details/24765

C19. Overall, M., Mukai, J., **Sakhare, R.S.**, Desai, J., Hillary, L., & Bullock, D. M. (2026, January). Frameworks for Connected Traffic and Emergency Operations Management – Poster Session. In Transportation Research Board 105th Annual Meeting (No. TRBAM-26-03559). TRB. https://annualmeeting.mytrb.org/OnlineProgram/Details/24533

C18. Mathew, J., Desai, J., **Sakhare, R.S.**, Hunter, J., & Bullock, D. M. (2025, January). Innovative Use of Emerging Data and Criteria for Maintenance Management – Lectern Session. In Transportation Research Board 104th Annual Meeting (No. TRBAM-25-01326). TRB. https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/22915

C17. Brinster, G., Desai, J., Overall, M., Gartner, C., **Sakhare, R.S.**, Mathew, J., Evans, N., & Bullock, D. M. (2025, January). Innovative Use of Emerging Data and Criteria for Maintenance Management – Lectern Session. In Transportation Research Board 104th Annual Meeting (No. TRBAM-25-02443). TRB. https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/22915

C16. **Sakhare, R.S.**, Desai, J., Mathew, J., & Bullock, D. M. (2025, January). Statewide and National Transportation Data Research – Poster Session. In Transportation Research Board 104th Annual Meeting (No. TRBAM-25-03615). TRB. https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/22967

C15. Bullock, D. M., & **Sakhare, R.S.** (2025, January). State Department of Transportation Research: High-Value Project Winners – Poster Session. In Transportation Research Board 104th Annual Meeting (No. P25-20182). TRB. https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/22727

C14. Saldivar-Carranza, E., **Sakhare, R.S.**, Desai, J., Mathew, J., Sivakumar, A. J., Mukai, J. & Bullock, D. M. (2025, January). Intelligent Transportation Systems in the Era of Connected vehicles and Vehicle to Everything Communication – Poster Session. In Transportation Research Board 104th Annual Meeting (No. TRBAM-25-03624). TRB. https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/22967

C13. Desai, J., Saldivar-Carranza, E., **Sakhare, R.S.**, Mathew, J., & Bullock, D. M. (2025, January). Toward Better Driving: Innovations for Improved Driving Performance and Enhanced Safety – Poster Session. In Transportation Research Board 104th Annual Meeting (No. TRBAM-25-02784). TRB. https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/22931

C12. Desai, J., Mathew, J., Li, H., **Sakhare, R.S.**, Horton, D., & Bullock, D. M. (2024, January). Statewide and National Data and Information Systems – Poster Session. In Transportation Research Board 103rd Annual Meeting (No. TRBAM-24-04421). TRB. https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/21103

C11. **Sakhare, R.S.**, Zhang, Y., Li, H., & Bullock, D. M. (2023, January). Road Weather Management Solutions for Improved Safety and Operations – Lectern Session. In Transportation Research Board 102nd Annual Meeting (No. TRBAM-23-02713). TRB. https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/19287

C10. Mahlberg, J., Desai, J., Li, H., **Sakhare, R.S.**, Wells, T., & Bullock, D. M. (2023, January). Advancements Available to Maintenance Equipment Fleets Within Departments of Transportation and Other Fleet Operations – Lectern Session. In Transportation Research Board 102nd Annual Meeting (No. TRBAM-23-01921). TRB. https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/19002

C9. **Sakhare, R.S.**, Hunter, M., Mukai, J., Li, H., & Bullock, D. M. (2023, January). Road Scholars: New Research in Travel Time, Speed, and Reliability Data – Poster Session. In Transportation Research Board 102nd Annual Meeting (No. TRBAM-23-01814). TRB. https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/19057

C8. Desai, J., Mahlberg, J., Woosung, K., **Sakhare, R.S.**, Li, H., McGuffey, J., & Bullock, D. M. (2022, January). Advancements in Winter Maintenance – Lectern Session. In Transportation Research Board 101st Annual Meeting (No. TRBAM-22-02228). TRB. https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/17477

C7. **Sakhare, R.S.**, Desai, J., Mahlberg, J., Mathew, J., Woosung, K., Li, H., McGregor, J., & Bullock, D. M. (2022, January). Freeway and Regional Transportation Systems and Operations – Poster Session. In Transportation Research Board 101st Annual Meeting (No. TRBAM-22-02130). TRB. https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/17526

C6. Mahlberg, J., **Sakhare, R.S.**, Li, H., Mathew, J., Bullock, D. M. & Surnilla, G. (2022, January). Traffic Control Devices – Poster Session. In Transportation Research Board 101st Annual Meeting (No. TRBAM-22-01292). TRB. https://annualmeeting.mytrb.org/OnlineProgramArchive/Details/17477

C5. Desai, J., **Sakhare, R. S.**, Rogers, S., Mathew, J. K., Habib, A., & Bullock, D. (2021, September). Using Connected Vehicle Data to Evaluate Impact of Secondary Crashes on Indiana Interstates. In 2021 IEEE International Intelligent Transportation Systems Conference (ITSC) (pp. 4057-4063). IEEE. https://doi.org/10.1109/ITSC48978.2021.9564653

C4. **Sakhare, R. S.**, Desai, J. C., Li, H., Mathew, J. K., & Bullock, D. M. (2021, January). Predicting Weather Impact on Traffic Speeds Using the Multi-Radar Multi-Sensor System. In 101st American Meteorological Society Annual Meeting. AMS. https://ams.confex.com/ams/101ANNUAL/meetingapp.cgi/Paper/382590

C3. **Sakhare, R. S.**, Mathew, J. K., Avr, A., Hubbard, S. M., Devi, L., & Bullock, D. M. (2018, January). Comparison of Bluetooth and Bus GPS Data for Estimating Arterial Travel Time and Trip Chaining. In Transportation Research Board 97th Annual Meeting (No. TRBAM-18-01695). TRB. https://trid.trb.org/View/1495014

C2. Li, H., **Sakhare, R. S.**, Mathew, J. K., Mackey, J., & Bullock, D. M. (2018, January). Estimating Intersection Control Delay Using High Fidelity Commercial Probe Vehicle Trajectory Data. In Transportation Research Board 97th Annual Meeting (No. TRBAM-18-01695). TRB. https://trid.trb.org/View/1494388

C1. McNamara, M., **Sakhare, R. S.**, Li, H., Baldwin, M., & Bullock, D. M. (2017, January). Integrating Crowdsourced Probe Vehicle Traffic Speeds into Winter Operations Performance Measures. In Transportation Research Board 96th Annual Meeting (No. TRBAM-17-00161). TRB. https://trid.trb.org/View/1437147

Key Technical Presentations

P9. **Sakhare, R.S.**, & Taylor, R. (2026, March). Segment Mobility and Signal Performance: Leveraging CV Metrics – Lectern Session. In 112th Purdue Road School (No. 131).

P8. **Sakhare, R.S.**, Thompson, A., & Sturdevant, N. (2026, March). Evaluation of Asset Identification Technology Utilizing Truck Images – Lectern Session. In 112th Purdue Road School (No. 159).

P7. **Sakhare, R.S.**, & McGregor, J. (2026, March). Utilizing CV Data and Truck Imagery for Roadway Monitoring – Lectern Session. In 112th Purdue Road School (No. 159).

P6. Gartner, C., & **Sakhare, R.S.** (2026, February). Connected Vehicle Data Use Cases for Practitioners – Lectern Session. In 2026 ITE Indiana Section February Tech Lunch. https://indianaite.starchapter.com/meetinginfo.php?id=202&ts=1768590344

P5. **Sakhare, R.S.**(2026, February). Work Zone Performance Measures Derived from Connected Vehicle Data that Map into the FHWA Rule on Work Zone Safety and Mobility – Seminar. In Purdue ITE Weekly Seminar Series.

P4. **Sakhare, R.S.**(2023, April). Impact of Rain Intensity on Interstate Traffic Speeds Using Connected Vehicle Data – Seminar. In Purdue ITE Weekly Seminar Series.

P3. **Sakhare, R.S.**, & Hillary, L. (2023, March). Identifying Traffic Incident Management Event Sequence using CV Data, Crash Reports, and ITS Images – Lectern Session. In 109th Purdue Road School (No. 61).

P2. **Sakhare, R.S.** (2023, March). Automation System Performance Issues/Truck and Car CV Penetration – Lectern Session. In 109th Purdue Road School (No. 117).

P1. **Sakhare, R.S.**, & Desai, J. (2023, March). Methodology for Monitoring Work Zones Traffic Operations using CV Data: Implementation on Indiana Interstate with UAS Imagery – Lectern Session. In 109th Purdue Road School (No. 182).

FUNDED RESEARCH PROJECTS

Roles - PI: Principal Investigator; RE: Research Engineer; GRA: Graduate Research Assistant

Funding Organizations - FHWA: Federal Highway Administration; INDOT: Indiana Department of Transportation; CCAT: Center for Connected and Automated Transportation.

- **FHWA —** TPF-5(514): Work Zone Analytics (RE, 2023-2026, $0.89M)
- **FHWA —** TPF-5(519): Expansion: Enhanced Traffic Signal Performance Measures (RE, 2023-2026, $1.3M)
- **INDOT —** SPR-5005: Evaluation of Asset Identification Technology (**Co-PI, 2025-2027**, $301,966)
- **INDOT —** SPR-5021: Traffic Signal Freight Prioritization via Vehicle to Infrastructure (V2I) Communications (**Co-PI, 2026-2027**, $383,966)

- **INDOT —** SPR-5020: Identifying Locations with Abnormally High Wrong Way Driving or Interstate U-Turns (**Co-PI, 2025-2027**, $224,109)
- **INDOT —** SPR-4928: Connected Vehicle Trajectory Data to Screen Network for Hard Braking and Hard Acceleration Events (**Co-PI, 2024-2025**, $218,390)
- **INDOT —** SPR-4933: Worksite Speed Control System (WSCS) (RE, 2024-2025)
- **INDOT —** SPR-4907: Systemwide Asset Condition Assessment using Connected Vehicle Data (RE, 2024-2025)
- **INDOT —** SPR-4857: Statewide Screening of Signalized Intersections for Capacity Improvements (RE, 2023-2024)
- **INDOT —** SPR-4854: MOT for MOT (RE, 2023-2025)
- **INDOT —** SPR-4851: Development of Incident Management Performance Measure Database and IN Time Training Material (RE, 2023-2025)
- **INDOT —** SPR-4850: Continued Deployment of Indiana Work Zone Analytics (RE, 2023-2025)
- **INDOT —** SPR-4803: Communication of Fixed and Mobile Warning to Commercial Trucks Using in Cab Notification (RE, 2023-2025)
- **INDOT —** SPR-4704: Evaluating the Robustness of MDSS Forecast and Compliance with Recommendations (RE, 2022-2024)
- **INDOT —** SPR-4639: Connected Vehicle Centric Dashboards for TMC of the Future (GRA, 2021-2023)
- **INDOT —** SPR-4636: Research Support to INDOT on I-465 Southeast Variable Speed Limit and Ramp Meter Project (RE, 2021-2025)
- **INDOT —** SPR-4603: Crowdsourcing/Winter Operations Dashboard Upgrade (GRA, 2021-2023)
- **INDOT —** SPR-4536: Implementation of Enhanced Probe Data (CANBUS) for Tactical Workzone and Winter Operations Management (GRA, 2020-2023)
- **CCAT —** Methodologies for Monitoring Work Zone Performance Measures across State Border Highways using Connected Vehicle Data (RE, 2023-2025)
- **INDOT —** SPR-4600: Impacts to Traffic Behavior from Queue Warning Truck – Current Pilot Project (GRA, 2021-2022)
- **INDOT —** SPR-4524: Evaluation of the Impact of On Vehicle Digital Communication Alerts to Improve INDOT and Motorist Safety (GRA, 2020-2022)
- **INDOT —** SPR-4451: Integration of Probe Data Tools into TMC Operations (GRA, 2020-2022)

TEACHING & MENTORING EXPERIENCE

- **Guest Lecture** for CE36100 Transportation Engineering, Purdue University on topic of *Connected Vehicle Data and its Applications (April 29, 2026)*
- **Invited Speaker** at the Transportation division, Department of Civil Engineering, IIT Madras on *Methodology for the Identification of Shock Wave Type and Speed in a Traffic Stream Using Connected Vehicle Data*; Interaction with undergraduate and graduate student in Civil Engineering. *(August 21, 2024)*
- **Instructor** for Mobility Course and Workshop SPARC (Scheme for Promotion of Academic and Research Collaboration, Government of Indian Initiative) - Developed course material and taught 3 session modules *(2021)*
- **Teaching Assistant** to Dr. Atul Narayan for CE2080 Civil Engineering Surveying course Department of Civil Engineering, IIT Madras *(2017-2018)*
- **Courses Prepared to Teach:** Transportation Engineering; Big Data Analytics in Transportation; Traffic Operations

Graduate Student Research Co-Advising/Mentorship

- **Myles Overall, Ph.D. in Civil Engineering —** Incident Management Performance Measures (SPR-4851)
- **Chris Gartner, M.S. in Civil Engineering —** Statewide performance measures (SPR-4857)
- **Andrew Thompson, M.S. in Electrical and Computer Engineering —**Road Asset Identification (SPR-4907)
- **Justin Mukai, M.S. in Civil Engineering —** Secondary crashes at the back of queue (J29)
- **Thomas Driscall, M.S. in Mechanical Engineering —** Cross-border queueing (J23)

- **Neelesh Gopalkrishnan, M.S. in Computer Science —** CV Penetration dashboard
- **Ahsmitha Sivakumar, B.Tech. in Civil Engineering —** Toll Plaza performance measures
- **Anirud Nandakumar, B.Tech. in Civil Engineering —** Normalized Hard braking evaluations

HONORS, AWARDS & ACHIEVEMENTS

- **2026 —** Received PRIME Grant from Purdue University in support of professional development
- **2024 —** Selected for the Google Cloud Research Innovator as part of the fourth cohort
- **2023 —** Edward J. Cox Memorial Scholarship, Indiana ITE Section
- **2023 —** Best Student Speaker Award, Purdue ITE
- **2021 —** ITS Midwest Project of the Year Award Winner
- **2021 & 2022 —** Elevator Pitch Event Winner, Purdue ITE
- **2021 —** Student Team Desing Competition Winner, Great Lakes District ITE
- **2020 —** International Collegiate Traffic Bowl Championship Winner, ITE International
- **2020 —** Best Poster Presentation Sigma Xi Scientific Society, Purdue University
- **2020-2023 —** Christopher B. & Susan S. Burke Graduate Program Research Assistantship, Purdue University
- **2018 —** Exemplary and All-round Best Performance in Dual Degree Program, Civil Engineering, Indian Institute of Technology (IIT) Madras, India
- **2018 —** Winner at CEA Technical Events (Case Study, Prabandha, Bon Auto and Aquanomics), IIT Madras, Chennai, India
- **2017 —** Invited Visiting Undergraduate Student at Purdue University
- **2016 —** Gold Medal in Civil Engineering, National Design and Research Forum, Institute of Engineers, India
- **2016 —** Mr. H. C. Verma Diamond Jubilee Awards, Indian Geotechnical Society (IGS), India
- **2016 —** Purdue Undergraduate Research Experience (PURE) Program Scholar; Among the top 3 Civil Engineering students chosen across India
- **2016 —** Represented India (Team of four) at Engineering Case Competition in Thailand organized by the Faculty of Chulalongkorn University
- **2015 —** Winner of Smart City Challenge Vellore-Engineers for Life Program
- **2015 —** Successfully Completed the Engineers for Life Program, AidVision Initiative
- **2014 —** Village's Choice Award, Faculty of Engineering, Chulalongkorn University, Thailand
- **2014 —** Represented India at the Engineering Case Competition (ECC) in Thailand
- **2013 —** Ranked top 0.2 percentile (2,445th) of 1.45 million students at IIT Joint Entrance Exam; considered amongst world's toughest examinations

ACADEMIC SERVICE

- **2025 Aug., - Current —** Transportation Research Board (TRB) **Committee Member** ACF13, Standing Committee on Limited Access Roadway Operations
- **2024, 2025 & 2026 — Planning Committee Member** Purdue Road School
- **2023, 2024 & 2025 — Session Moderator** Safety track (2023) Project Management/Planning track (2024) Project Management/Planning track (2025) at Purdue Road School
- **2022 —** Institute of Transportation Engineers (ITE) **Executive Board Member – Purdue Chapter;** Received *ITE Student Chapter Award* from ITE International; Received *Outstanding Student Chapter from ITE Great Lakes*
- **2024 —** Judge for Spring Undergraduate Research Conference
- Active Reviewer for following Journals:
    - Transportation Research Part C: Emerging Technologies
    - Journal of Transportation Engineering, Part A: Systems, ASCE
    - Transportation Letters: the International Journal of Transportation Research

- Transportation Research Borad
- Transportation Research Records
- Transportation Research Interdisciplinary Perspectives
- Journal of the Institution of Engineers (India): Series A
- Transportation in Developing Economies

COMMUNITY LEADERSHIP & EXTRACURRICULARS

- **2022 —** National Collegiate Cricket Association's (NCCA) Midwest Zone Championship Winner (Captain of the Purdue Cricket team)
- **2022 —** Midwest Cricket Tournament Runner-up (more than 60 participating teams)
- **2022 & 2023 —** Purdue Indoor Cricket League Winner (Captain of the team in 2023)
- **2018 —** Winner in CEA Cricket Tennis Ball Tournament (runner up in 2017)
- **2017 —** Fourth in Men's Cricket, IITM Institute Gymkhana Intramural Competitions
- **2016 —** Head Member at International & Alumni Relations Student Council, IIT Madras
- **2016 —** Volunteer for Annual Terry Fox Run, The Terry Fox Foundation, Chennai
- **2015 —** Head Member at Civil Engineering Association (CEA), IIT Madras *(2015)*
- **2015 —** Student Mentor, MITR Office of the Dean Students, IIT Madras
- **2015 —** Ambience & Design Super-Coordinator at Shaastra, Technical Event, IIT Madras
- **2014 —** Events & Projects Coordinator at Sustainability Network, IIT Madras
- **2010 —** Competed at State Level Archery Competitions, Maharashtra, India

SELECT MEDIA COVERAGE

- **2026, Apr. 24** — USDOT ITS JPO (2026-L01279), Use Connected Vehicle Data to Analyze Cross-Border Queuing, Detect Emerging Issues, and Identify Traffic Trends. https://www.itskrs.its.dot.gov/2026-l01279
- **2025, Dec. 01** — USDOT ITS JPO, Vehicle-to-Everything (V2X) Communications. https://www.itskrs.its.dot.gov/success-strategies/snapshot/vehicle-everything-v2x-communications
- **2024, Dec. 24** — The New York Times, How Your Car Might Be Making Roads Safer. https://www.nytimes.com/2024/12/20/technology/connected-cars-roads-data.html
- **2023, Apr. 30** — USDOT ITS JPO (2023-B01745), Indiana DOT's 26 Month Pilot Study Shows That Deployment of Queue Warning Trucks Ahead of Interstate Work Zones to Alert Motorists of Queued Traffic Reduced Hard Braking Events by 80 Percent. https://www.itskrs.its.dot.gov/2023-b01745
- **2023, Mar. 22** — Purdue News, Connected vehicles the latest tool to give engineers real-time insight into highway traffic congestion issues. https://www.purdue.edu/newsroom/2023/Q1/connected-vehicles-the-latest-tool-to-give-engineers-real-time-insight-into-highway-traffic-congestion-issues
- **2022, Mar. 25 — USDOT ITS JPO** (2022-L01100), Place Queue Trucks Ahead of a Work Zone to Reduce Number of Hard Braking Events and Crash Risks. https://www.itskrs.its.dot.gov/2022-l01100
- **2021, Apr. 07** — Google Cloud Blog, Purdue University traffic research program cuts data analysis and batching from hours to minutes with BigQuery. https://cloud.google.com/customers/purduejtrp
- **2021, Sept. 17** — HAAS Alert *Industry News*, Digital Alerting Helps Reduce Hard Braking by 80% in Purdue University Study. https://www.haasalert.com/report/purdue-study

SCHOLARLY PROFILES

- **Google Scholar:** https://scholar.google.com/citations?user=4crwCDoAAAAJ&hl=en
- **Research Gate:** https://www.researchgate.net/profile/Rahul-Suryakant-Sakhare
- **ORCiD:** https://orcid.org/0000-0001-7843-5707
- **LinkedIn:** https://www.linkedin.com/in/rahulsakhare/